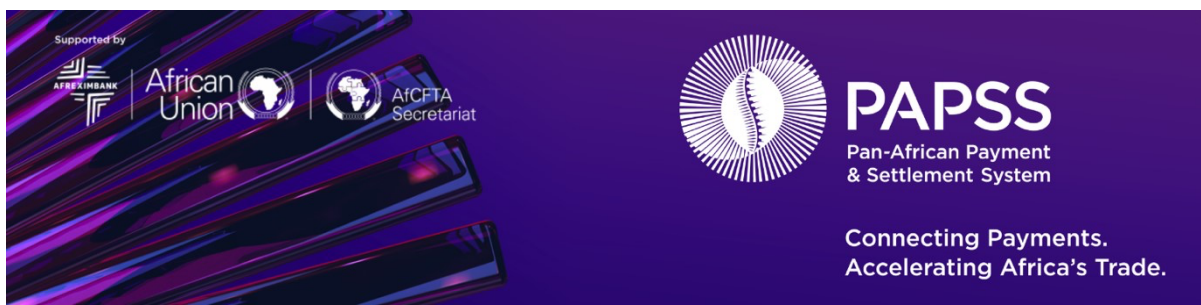


VERSION: 1.05



REST API Middleware

PAPSS TECHNICAL INTEGRATION

PAPSS TECHNOLOGY TEAM

PAPSS.COM

Cairo, Egypt

This item is classified as P rivate

This item is classified as Public

Document History

DATE	VERSION	CHANGES
2023-02-12	1.01	First Draft
2023-05-15	1.02	Update REST API Specification
2023-07-18	1.03	Update Test Cases and Error Messages
2023-10-10	1.04	Updated API Payloads
2024-02-15	1.05	Updated API Payloads
2024-12-20	1.06	FX Rate Update

This item is classified as P rivate

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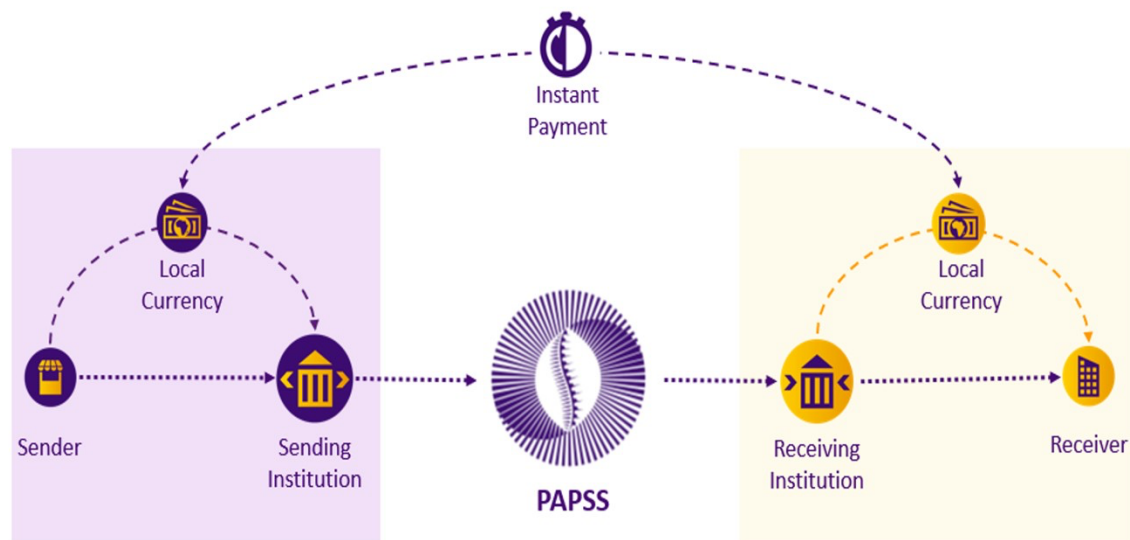
1. Getting Started with PAPSS Integration

1.1. Introduction

PAPSS (Pan-African Payment and Settlement System) - is a cross-border, financial market infrastructure enabling payment transactions across Africa.

With PAPSS APIs, you can make instant and secure cross boarder payments in Local and USD currencies across Africa depending on the Settlement Model that has been chosen by the bank.

The information exchange between the different parties is based on **ISO-20022** standard.



1.2. Integrating to PAPSS Instant Payments System

To use integrate with PAPSS, you will need to be an PAPSS customer and participant and setup on the PAPSS Instant Payment System. If you are new to PAPSS and interested in becoming a client, contact us at info@papss.com to learn more.

Integration is achieved using either of 2 methods documented below:

1.2.1. Integrating through the PAPSS REST API Middleware

Integrating the PAPSS REST API middleware is the focus of this document. For this integration to be achieved a server connected to the PAPSS VPN will have to be provisioned in the participant's environment.

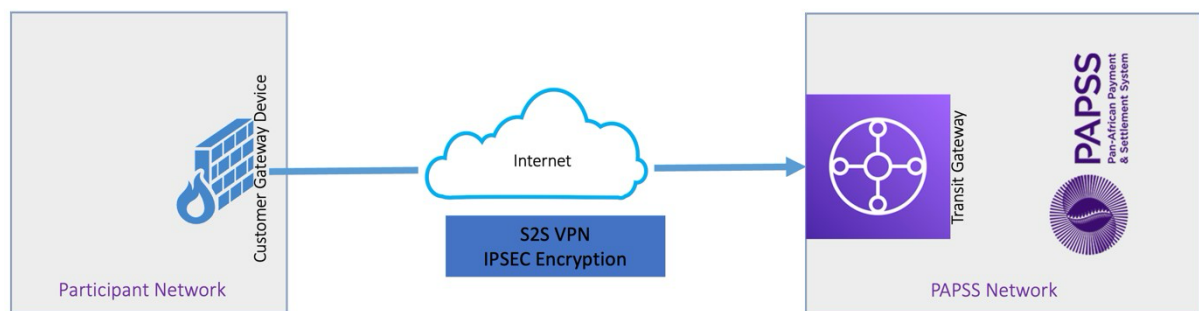
1.2.2. Integrating through the PAPSS Java Client Library

To integrate using the Java Client Library, head over to the online documentation here:

This item is classified as Private <https://papssfrica.github.io>

1.3. Integration Requirements

1.3.1. VPN



Participants

- A customer gateway device is a physical or software appliance that a participant own or manage in their on-premises network

PAPSS

- AWS Transit Gateway connects Internal(PAPSS) and External (Participants) networks through a central hub. It acts as a cloud router.

VPN Connection

- **IPSEC VPN** is established between Participants and PAPSS Network
- Both **Static** and **BGP tunnels** are supported

1.3.2. Digital Certificate

The security mechanism used by PAPSS Instant Payment System guarantees confidentiality, integrity and authenticity of the data transferred between the participant systems and the IPS.

Technically, the PAPSS IPS has two security levels:

1. SSL technology for the encryption of the communication channel between the Participants system and the central PAPSS IPS.
2. Digital signature on each ISO 20022 message sent to the PAPSS Instant Payment System.

Each level of security requires a different digital certificate.

The digital certificate must be uploaded into the central IPS. The private key of each digital certificate must be kept only at the Participant and never be share with anyone to ensure the security and integrity of the communication

1.3.3. Participant Development Team

The participant's software development team will be responsible to integrating PAPSS payment flows and operations to the participants core banking systems for both inwards and outwards operations.

In the case of a commercial bank, the participants software development team will also be responsible for developing logic to expose PAPSS to the bank's front-end interfaces / channels that consumers of PAPSS will interact with to initiate payments.

Some of the channel considerations include but are not limited to:

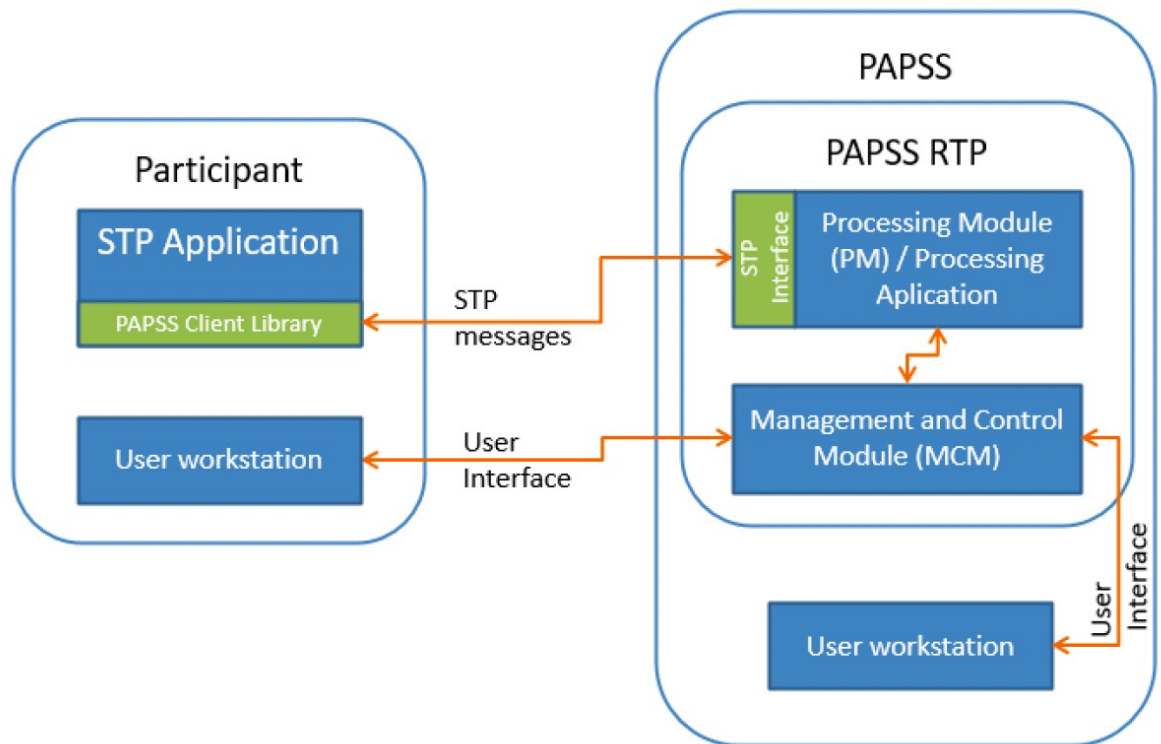
1. Trade systems
2. Mobile app
3. Branch banking
4. Online banking

The software development team will also develop pricing features into their systems for participant fee collection in accordance with the pricing guidelines that have been shared with business.

1.3.4. Server / Hardware Requirements

	Test Environment	Production Environment
<i>Preferred Operating System</i>	Red Hat Enterprise Linux 8.8.0 or above	Red Hat Enterprise Linux 8.8.0 or above
<i>Alternative Operating System</i>	Windows Server 2019 or above	Windows Server 2019 or above
<i>Processors</i>	8 Cores	16 Cores
<i>Memory</i>	8 GB RAM	16 GB RAM
<i>Hard Disk Capacity</i>	250 GB Minimum	250 GB Minimum
<i>Runtime Technology</i>	OpenJDK Runtime Environment 17	OpenJDK Runtime Environment 17
<i>Network Connectivity</i>	Internet Access, PAPSS VPN	Internet Access, PAPSS VPN

1.4. Architecture



2. PAPSS Operations

2.1 Introduction

Integrated solutions for participants will have the following scope of features:

- Outwards Operations (Participant acting as Originator of Transactions)
 - Send and process Credit Transfer (pacs.008.001)
 - Send and process Name/Account Inquiry Requests (acmt.023.001)
 - Send and process Payment Status Requests (pacs.028.001)
 - Send and process Payment Recall Messages (camt.056.007)
 - Send Participant Status Requests
 - Send FX Rate Requests
 - Send Get Positions Requests
- Inwards Operations (Participant acting as Beneficiary of Transactions)
 - Receive and process Credit Transfer Requests (pacs.008.001)
 - Receive and process Name/Account Inquiry Requests (acmt.023.001)
 - Receive and process Settlement Recon Messages (rcon.001)
 - Receive and process Payment Recall Messages (camt.056.007)
 - Positive Reply to Payment Recall / Credit Return (pacs.004)
 - Negative Reply to Payment Recall (camt.029.001)

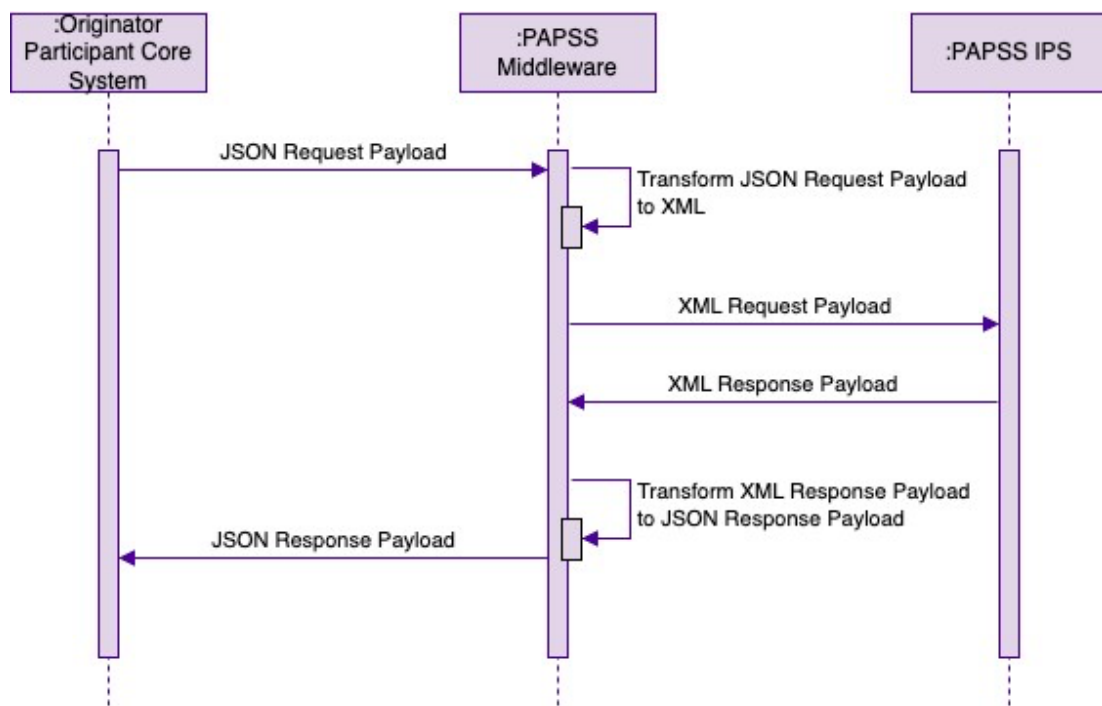
2.2 Outwards Operations – Sending Messages

Sending messages to PAPSS IPS is achieved through a HTTP REST APIs in the PAPSS-OutwardsMiddleware-Service, whereby each outwards operation request is sent to a specific REST endpoint.

PAPSS IPS will process each request message and send back a response.

The Originator Participant System can send multiple such requests in parallel.

Each operation is synchronous, under normal operation conditions, for each request made to PAPSS IPS there will be a response message.



The following outwards transactional and non-transactional operations are supported by the PAPSS IPS through the PAPSS-Outwards-Middleware-Service:

The following diagram depicts the flow of messages when initiating a Credit Transfer Request:

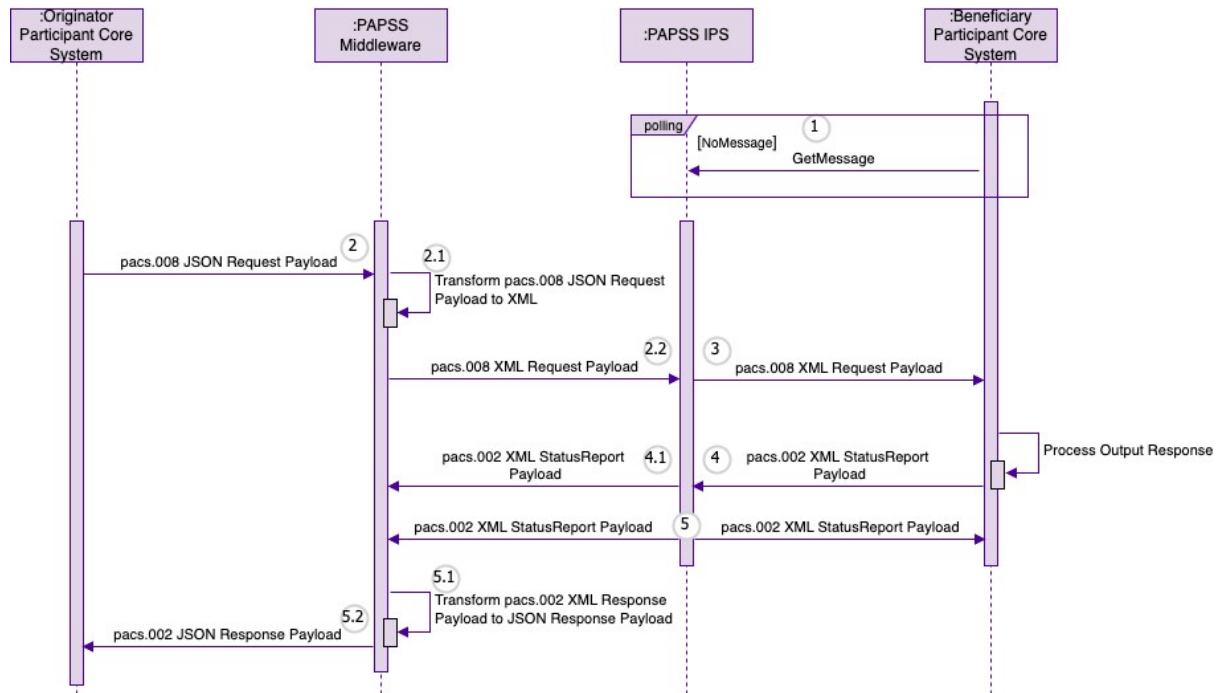


Figure 2: Message flow for Credit Transfer Request

2.3 Inwards Operations – Receiving Messages

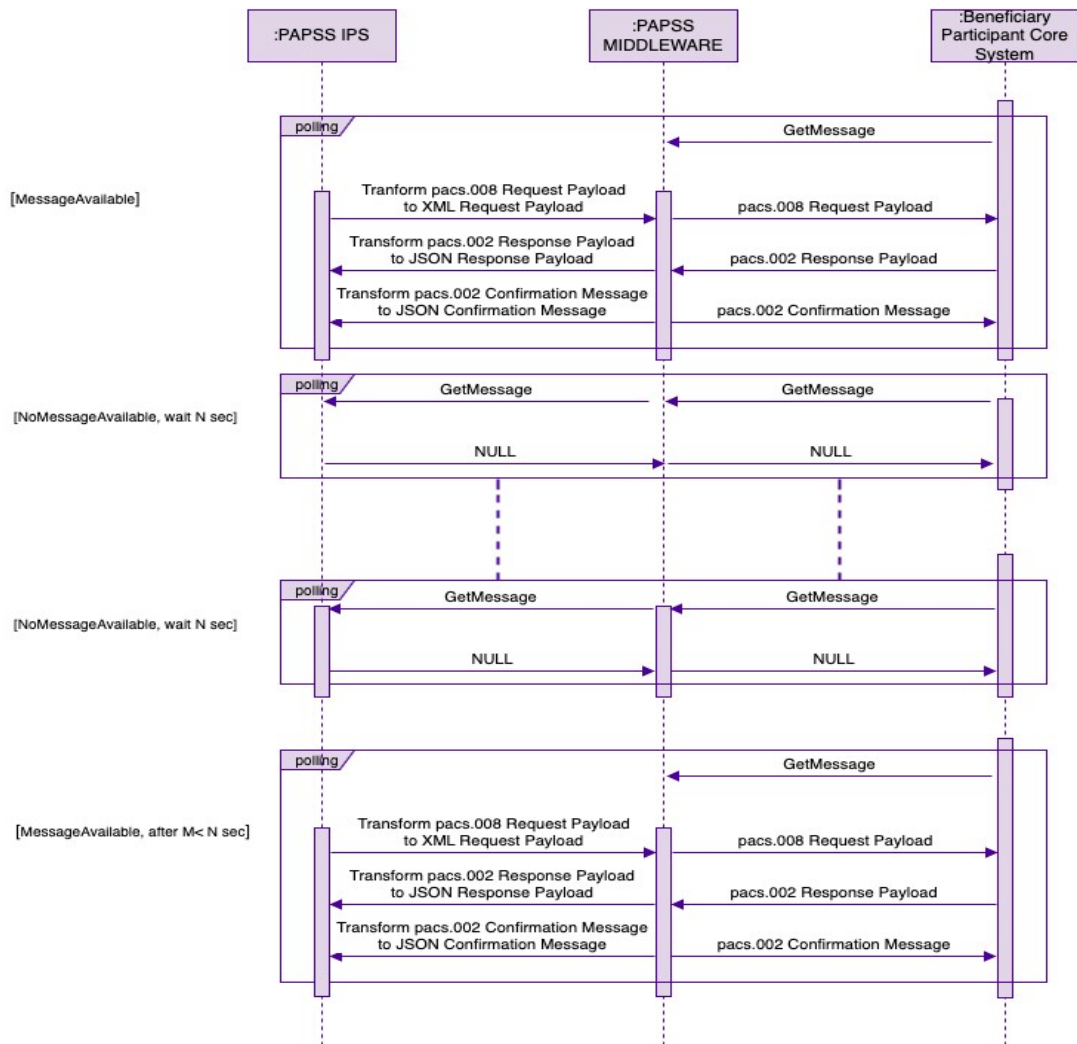
Inwards Operations are achieved through the PAPSS-Inwards-Middleware-Service that integrates with the PAPSS IPS to receive messages through a HTTP Long-Polling mechanism and then integrates with downstream Participant Core Systems for processing of messages.

This HTTP Long-Polling mechanism that the PAPSS-Inwards-Middleware-Service performs on behalf of the participant achieves 2 elements.

1. The HTTP Long-Polling mechanism receives messages from the IPS at configurable intervals (*Optimally 5 seconds*)
2. HTTP Long-Polling mechanism registers the participant as having an ONLINE status on the PAPSS IPS. In the context of PAPSS 24 x 7 operating model this is important as the participant must be permanently connected and ONLINE to process messages. An OFFLINE participant will not be able to receive messages.

Messages that are received by the PAPSS-Inwards-Middleware-Service are then stored locally and posted downstream to participant's Core Banking Systems using HTTP REST APIs that have been developed by the participants to process these incoming messages.

The Inwards REST API Specification is as defined in section 4.3 of this document.

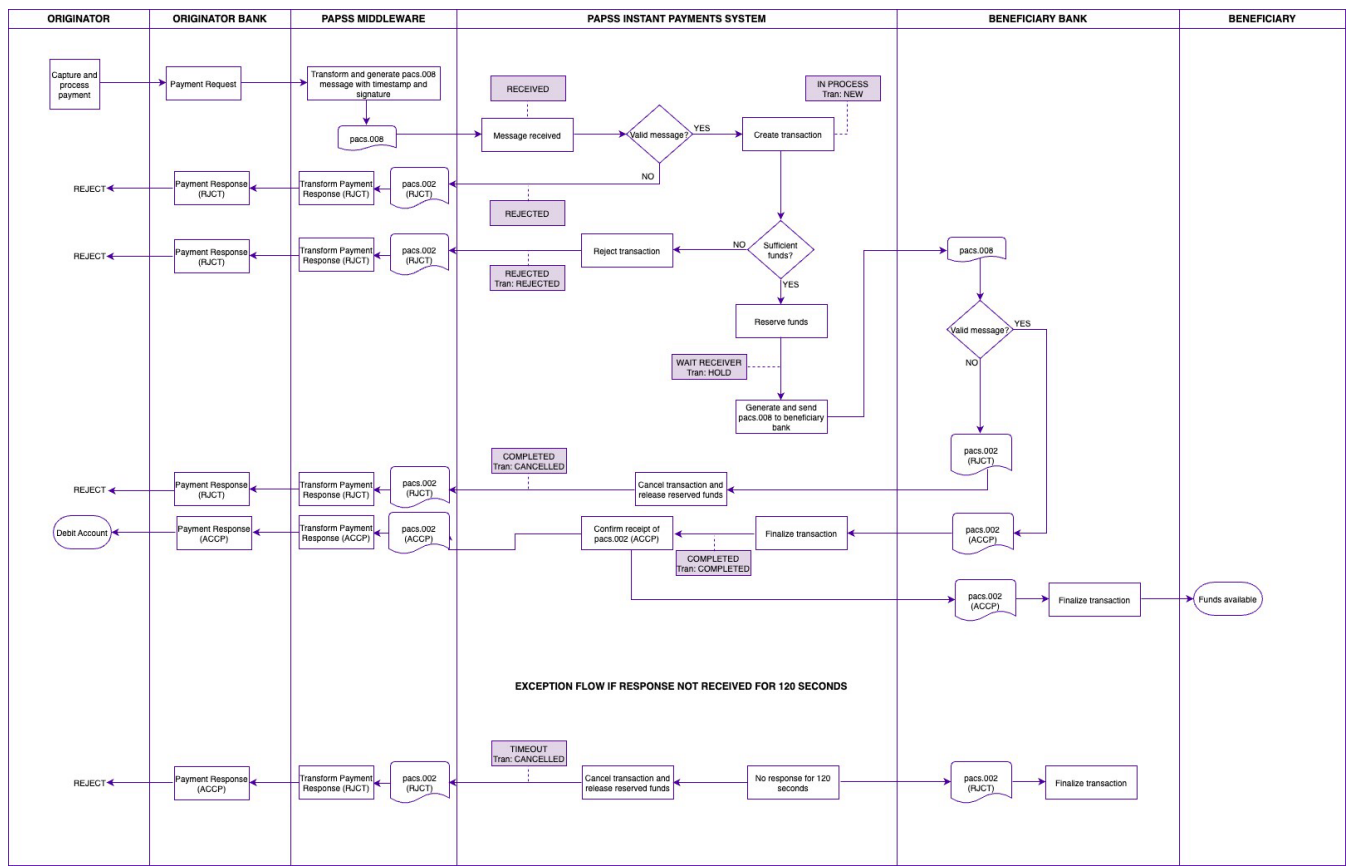


3. Message Processing

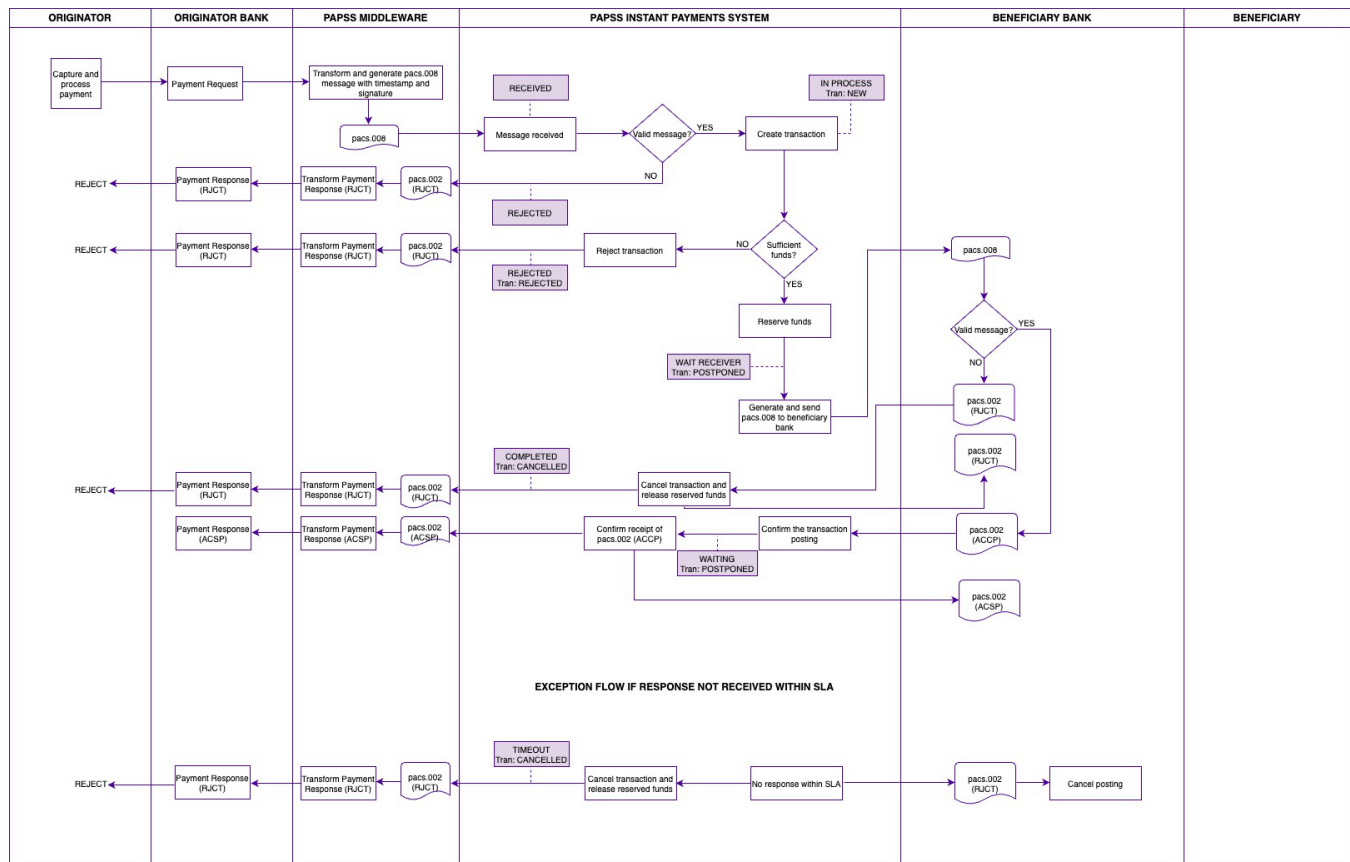
3.1. Name Enquiry Operation

This operation allows the Originating Participant to inquire and verify the validity of the Beneficiary account details before initiating a Credit Transfer. Name Enquiry operation affords the customer/teller the opportunity of comparing intended beneficiary name with the account name tied to the provided beneficiary account in the destination bank, before confirming a transfer. So, the sender bank makes a request with the account number specified in the request message and the name corresponding to the account number is supplied in the response message provided by the destination bank.

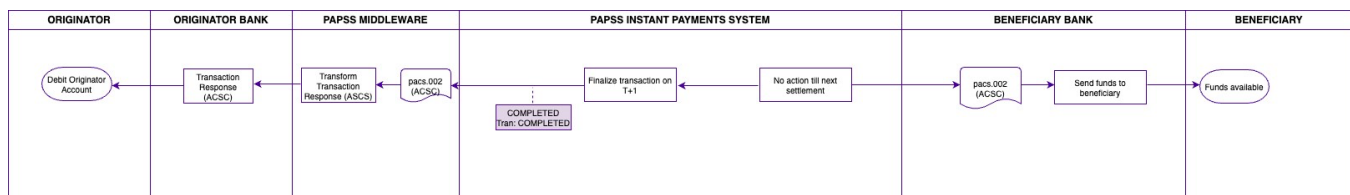
3.2. Credit Transfer Request Operation in Instant Payments Region



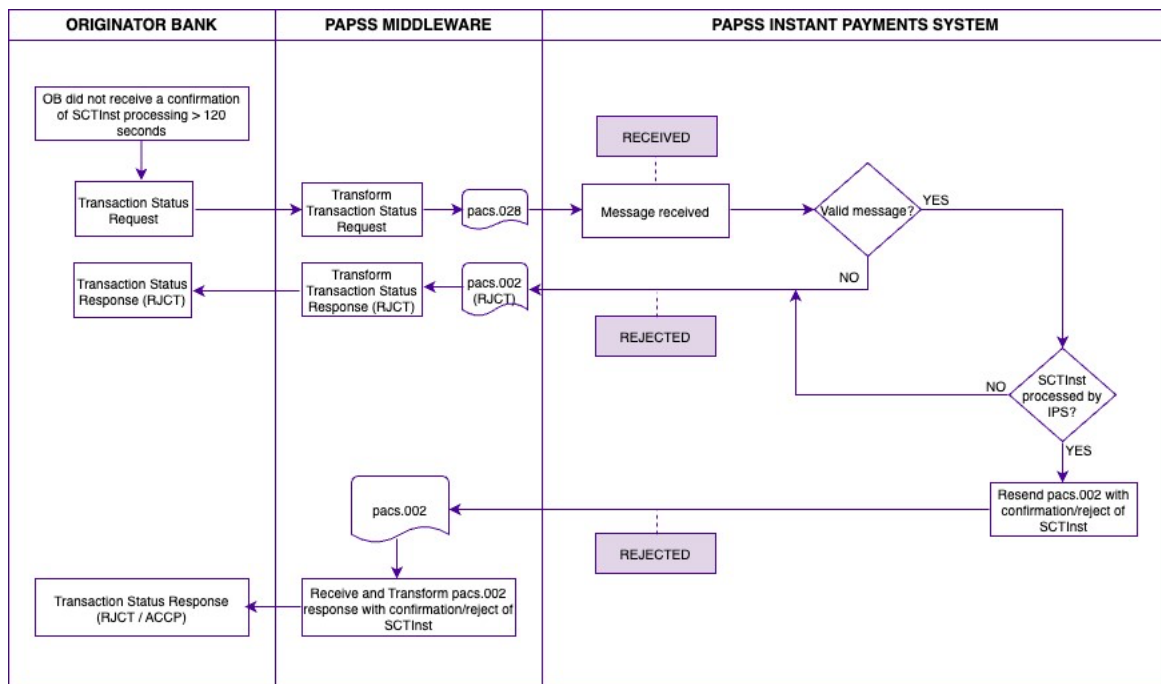
3.3. Credit Transfer Request Operation in Non-Instant Payments Region



Finalizing Credit Transfer Operation at T+1 for postponed transactions.



3.4. Transaction Status Inquiry Operation



4. REST API Specification

This item is classified as P rivate

4.1. Introduction

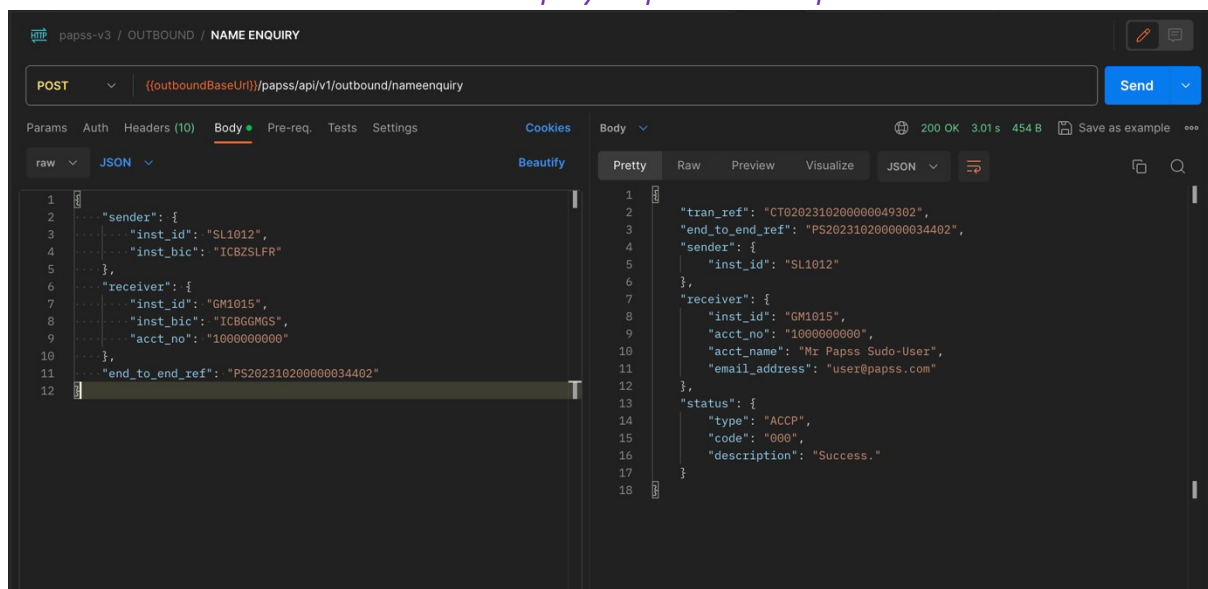
REST APIs in the PAPSS REST API Middleware are categorized into 2 types:

1. **Outbound Operations REST APIs** - These are defined as APIs in which the participant initiates the call to the API endpoints with a HTTP Request payload and a response is synchronously expected from the same endpoint.
2. **Inbound Operations REST APIs** – These are defined as API endpoints that will be developed by the participant in its core banking system and to which messages will be pushed by the middleware to these API endpoints for processing by the Participant's Core Banking System. Responses from the participant's systems are expected in a synchronous manner.

4.2. Outbound Operations - REST API Specification

4.2.1. Name Inquiry REST API Specification

4.2.1.1. Outbound Name Inquiry Request and Response



4.2.1.2. Outbound Name Inquiry Request Payload

```
{
  "sender": {
    "inst_id": "SL1012",
    "inst_bic": "ICBZSLFR"
  },
  "receiver": {
    "inst_id": "GM1015",
    "inst_bic": "ICBGGMGS",
    "acct_no": "1000000000"
  },
  "end_to_end_ref": "PS202310200000034402"
}
```

4.2.1.3 Outbound Name Inquiry Response Payload

```
{
  "tran_ref": "CT0202310200000049302",
  "end_to_end_ref": "PS202310200000034402",
```

This item is c

```

"sender": {
  "inst_id": "SL1012"
},
"receiver": {
  "inst_id": "GM1015",
  "acct_no": "1000000000",
  "acct_name": "Mr Papss Sudo-User",
  "email_address": "user@papss.com"
},
"status": {
  "type": "ACCP",
  "code": "000",
  "description": "Success."
} }

```

4.2.2 Credit Transfer REST API Specification

4.2.2.1 Outbound Credit Transfer Request and Response

The screenshot displays a REST client interface for the endpoint `POST {{outboundBaseUrl}}/papss/api/v1/outbound/fundtransfercredit`. The request body is a JSON object with the following structure:

```

{
  "sender": {
    "inst_id": "SL1012",
    "inst_bic": "ICBZSLFR",
    "ctry": "SL",
    "ccy": "SLE",
    "acct_no": "1000000000",
    "acct_name": "John Doe",
    "address": "Tema"
  },
  "receiver": {
    "inst_id": "GM1015",
    "inst_bic": "ICBGGMGS",
    "ctry": "GM",
    "ccy": "GMD",
    "acct_no": "435000100",
    "acct_name": "Mr Papss Sudo-User",
    "address": "user@papss.com"
  },
  "amount": 100,
  "is_invoice": false,
  "narration": "Trade Transfer Invoice for purchase of Cassava",
  "tran_date_time": "2023-01-13T12:08:18.344Z",
  "end_to_end_ref": "ENDTOENDREF123",
  "ctgy_purp": "SUPP",
  "lc_instmt_type": "ET"
}

```

The response is a 200 OK status with a response time of 2.31s and a body size of 475 B. The response body is a JSON object:

```

{
  "tran_ref": "PS202310200000050302",
  "tran_date_time": "2023-10-20T14:12:18.586Z",
  "orgnl_txn_info": {
    "tran_ref": "20231020SL1012141218",
    "end_to_end_ref": "ENDTOENDREF123",
    "status": {
      "type": "RJCT",
      "reason": "AM23"
    }
  },
  "status": {
    "type": "AM23",
    "code": "2000",
    "description": "Insufficient funds for the transaction clearing."
  }
}

```

4.2.2.2 Outbound Credit Transfer Request

```
{
  "sender": {
    "inst_id": "SL1012",
    "inst_bic": "ICBZSLFR",
    "ctry": "SL",
    "ccy": "SLE",
    "acct_no": "1000000000",
    "acct_name": "John Doe",
    "address": "Tema"
  },
  "receiver": {
    "inst_id": "GM1015",
    "inst_bic": "ICBGGMGS",
    "ctry": "GM",
    "ccy": "GMD",
    "acct_no": "435000100",
    "acct_name": "Mr Papss Sudo-User",
    "address": "user@papss.com"
  },
  "amount": 100,
  "is_invoice": false,
  "narration": "Trade Transfer Invoice for purchase of Cassava",
  "tran_date_time": "2023-01-13T12:08:18.344Z",
  "end_to_end_ref": "ENDTOENDREF123",
  "ctgy_purp": "SUPP",
  "lc_instrmt_type": "ET" }
```

4.2.2.3 Outbound Credit Transfer Response

```
5  {
6    "tran_ref": "PS202310200000050302",
7    "tran_date_time": "2023-10-20T14:12:18.586Z",
8    "orgnl_txn_info": {
9      "tran_ref": "20231020SL1012141218",
10     "end_to_end_ref": "ENDTOENDREF123",
11     "status": {
12       "type": "RJCT",
13       "reason": "AM23"
14     }
15   },
16   "status": {
17     "type": "AM23",
18     "code": "2000",
19     "description": "Insufficient funds for the transaction clearing."
20   }
21 }
```

4.2.3 FX Rate Inquiry REST API Specification

4.2.3.1 FX Rate Inquiry Request & Response

The screenshot displays a REST client interface for the 'papss-v3 / OUTBOUND / FX RATE ENQUIRY' endpoint. The request is a POST to '{(outboundBaseUrl)}/papss/api/v1/outbound/rateenquiry'. The request body is a JSON object with the following structure:

```
1 {
2   "amount": 500,
3   "is_invoice": true,
4   "lc_instrmt_type": "SSA3",
5   "sender": {
6     "inst_id": "GM1015",
7     "ctry": "GM",
8     "ccy": "GMD"
9   },
10  "receiver": {
11    "inst_id": "GH1005",
12    "ctry": "GH",
13    "ccy": "GHS"
14  }
15 }
```

The response is a JSON object with the following structure:

```
1 {
2   "time": "2023-10-20T14:19:03Z",
3   "usd_to_sender_currency_rate": 300.0000,
4   "usd_to_receiver_currency_rate": 200.0000,
5   "sender_to_receiver_currency_rate": 0.6666666666,
6   "sender_amount": 750.00,
7   "receiver_amount": 500.00,
8   "usd_exchange_amount": 2.50,
9   "sender_currency_fee_amount": 600.00,
10  "usd_fee_amount": 2.00
11 }
```

4.2.3.2 FX Rate Inquiry Request Payload

```
{
  "amount": 500,
  "is_invoice": true,
  "lc_instrmt_type": "SSA3",
  "sender": {
    "inst_id": "GM1015",
    "ctry": "GM",
    "ccy": "GMD"
  },
  "receiver": {
    "inst_id": "GH1005",
    "ctry": "GH",
    "ccy": "GHS"
  }
}
```

4.2.3.3 FX Rate Inquiry Response Payload

```
{
  "time": "2023-10-20T14:19:03Z",
  "usd_to_sender_currency_rate": 300.0000,
  "usd_to_receiver_currency_rate": 200.0000,
  "sender_to_receiver_currency_rate": 0.6666666666,
  "sender_amount": 750.00,
  "receiver_amount": 500.00,
  "usd_exchange_amount": 2.50,
  "sender_currency_fee_amount": 600.00,
  "usd_fee_amount": 2.00
}
```

4.2.3.4 Update/SET FX Rate API Specification

Sample Request.

```
{
  "instId": "KE1002",
  "accountType": "SSA3",
  "buyRate": 120.50,
  "sellRate": 130.50,
  "countryCode": "KE",
  "localCurrency": "KES",
  "fxRateName": "KE1002-SSA_3-001-KES-USD"
}
```

Sample Response

```
{
  "tranRef": "20241220KE1002101547",
  "tranDateTime": "2024-12-20T09:15:47.325Z",
  "fxRateName": "KE1002-SSA_3-001-KES-USD",
  "status": {
    "type": "ACCP",
    "code": "000",
    "description": "Success."
  }
}
```

The screenshot displays a REST client interface with the following details:

- URL:** `{{baseUrl}}/papss-outbound-service/papss/api/v3/outbound/fx-rate-update`
- Method:** POST
- Body (raw):**

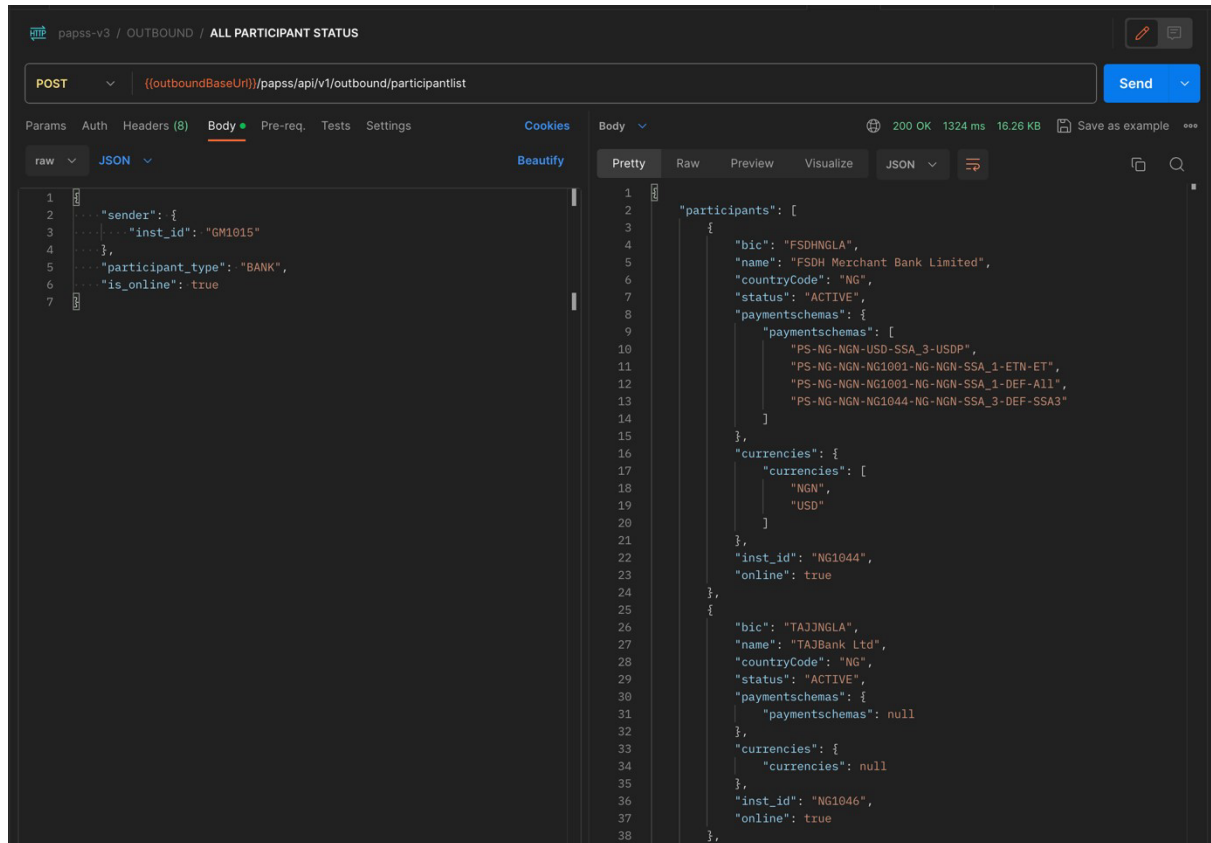
```
{
  "instId": "KE1002",
  "accountType": "SSA3",
  "buyRate": 120.50,
  "sellRate": 130.50,
  "countryCode": "KE",
  "localCurrency": "KES",
  "fxRateName": "KE1002-SSA_3-001-KES-USD"
}
```
- Response (JSON):**

```
{
  "tranRef": "20241220KE1002101547",
  "tranDateTime": "2024-12-20T09:15:47.325Z",
  "fxRateName": "KE1002-SSA_3-001-KES-USD",
  "status": {
    "type": "ACCP",
    "code": "000",
    "description": "Success."
  }
}
```
- Status:** 200 OK
- Time:** 5.59 s
- Size:** 343 B

This item is

4.2.4 Participant List REST API Specification

4.2.4.1 Participant List Request & Response



4.2.4.2 Participant List Request Payload

```
{  "sender": {    "inst_id": "GM1015"  },  "participant_type": "BANK",  "is_online": true}
```

4.2.4.3 Participant List Response Payload

```
{  "participants": [    {      "bic": "FSDHNGLA",      "name": "FSDH Merchant Bank Limited",      "countryCode": "NG",      "status": "ACTIVE",      "paymentschemas": {        "paymentschemas": [          "PS-NG-NGN-USD-SSA_3-USDP",
```

This item is deprecated.

```

        "PS-NG-NGN-NG1001-NG-NGN-SSA_1-ETN-ET",
        "PS-NG-NGN-NG1001-NG-NGN-SSA_1-DEF-All",
        "PS-NG-NGN-NG1044-NG-NGN-SSA_3-DEF-SSA3"
    ]
},
"currencies": {
    "currencies": [
        "NGN",
        "USD"
    ]
},
"inst_id": "NG1044",
"online": true
},
{
    "bic": "TAJJNGLA",
    "name": "TAJBank Ltd",
    "countryCode": "NG",
    "status": "ACTIVE",
    "paymentschemas": {
        "paymentschemas": null
    },
    "currencies": {
        "currencies": null
    },
    "inst_id": "NG1046",
    "online": true
}
] }

```

4.2.5 Transaction Status REST API Specification

4.2.5.1 Transaction Status Request & Response

The screenshot displays a REST client interface for a 'TRANSACTION STATUS QUERY' endpoint. The request is a POST to `{{(outboundBaseUrl)}/papss/api/v1/outbound/txnstatusquery}`. The request body is a JSON object with the following structure:

```

{
  "sender": {
    "inst_id": "SL1012"
  },
  "tran_ref": "20231020SL1012141225",
  "orgnl_txn_info": {
    "tran_ref": "20231020SL1012141218",
    "tran_date_time": "2023-10-20T14:12:18.586Z"
  }
}

```

The response is a 200 OK status with a JSON body containing transaction details:

```

{
  "tran_ref": "PS20231020000050302",
  "tran_date_time": "2023-10-20T14:35:25.569Z",
  "orgnl_txn_info": {
    "tran_ref": "20231020SL1012141218",
    "end_to_end_ref": null,
    "status": {
      "type": "RJCT",
      "reason": "AM23"
    }
  },
  "status": {
    "type": " ",
    "code": "000",
    "description": "Success."
  }
}

```

This item is cl

4.2.5.2 Transaction Status Request

```
{
  "sender": {
    "inst_id": "SL1012"
  },
  "tran_ref": "20231020SL1012141225",
  "orgnl_txn_info": {
    "tran_ref": "20231020SL1012141218",
    "tran_date_time": "2023-10-20T14:12:18.586Z"
  }
}
```

4.2.5.3 Transaction Status Response

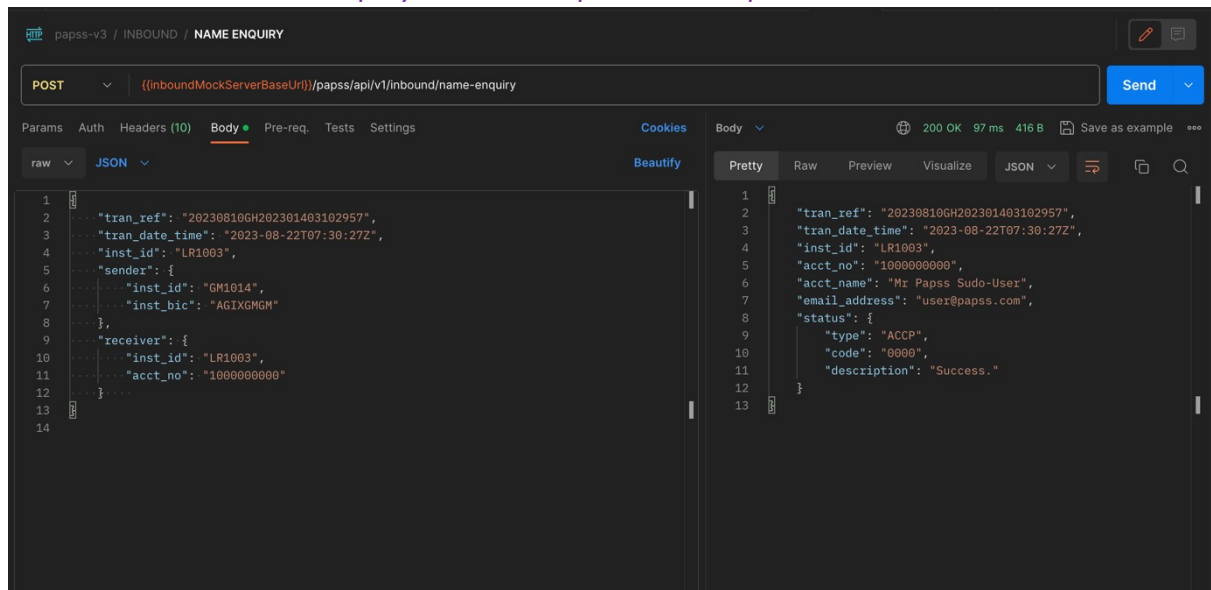
```
{
  "tran_ref": "PS202310200000050302",
  "tran_date_time": "2023-10-20T14:35:25.569Z",
  "orgnl_txn_info": {
    "tran_ref": "20231020SL1012141218",
    "end_to_end_ref": null,
    "status": {
      "type": "RJCT",
      "reason": "AM23"
    }
  },
  "status": {
    "type": " ",
    "code": "000",
    "description": "Success."
  }
}
```

4.3 Inbound Operations - REST API Specification

This item is classified as P ivate

4.3.3 Name Inquiry REST API Specification

4.3.3.1 Name Inquiry Inbound Request and Response



4.3.1.2 Name Inquiry Inbound Request Payload to Core Banking System

```
{  "tran_ref": "20230810GH202301403102957",  "tran_date_time": "2023-08-22T07:30:27Z",  "inst_id": "LR1003",  "sender": {    "inst_id": "GM1014",    "inst_bic": "AGIXGMGM"  },  "receiver": {    "inst_id": "LR1003",    "acct_no": "1000000000"  }}
```

4.3.1.3 Name Inquiry Inbound Response Payload from Core Banking System

```
{  "tran_ref": "20230810GH202301403102957",  "tran_date_time": "2023-08-22T07:30:27Z",  "inst_id": "LR1003",  "acct_no": "1000000000",  "acct_name": "Mr Papss Sudo-User",  "email_address": "user@papss.com",  "status": {    "type": "ACCP",    "code": "0000",    "description": "Success."  }}
```

4.3.4 Credit Transfer REST API Specification

4.3.4.1 Credit Transfer Inbound Request and Response

The screenshot shows a REST client interface with a POST request to `[[inboundMockServerBaseUrl]]/papss/api/v1/inbound/credit-transfer`. The request body is a JSON object with the following fields:

```
1 {
2   "tran_ref": "CT0202310040000049501",
3   "tran_date_time": "2023-10-04T14:43:18.000Z",
4   "receiver": {
5     "amount": 500.00,
6     "inst_bic": "ICBGGMGS",
7     "ctry": "GM",
8     "ccy": "GMD",
9     "acct_no": "1000000000",
10    "inst_id": "GM1015",
11    "acct_name": "Mr Papss Sudo-User",
12    "addr": "user@papss.com"
13  },
14  "sender": {
15    "amount": 253566.50,
16    "inst_bic": "ICBZSLFR",
17    "ctry": "SL",
18    "ccy": "SLE",
19    "acct_no": "435000100",
20    "inst_id": "SL1012",
21    "acct_name": "John Doe",
22    "addr": "Tema"
23  },
24  "ctgy_purp": "SUPP",
25  "narration": "Trade Transfer Invoice for purchase of Cassava",
26  "end_to_end_tran_ref": "20231004SL1012144316",
27  "lc_instrmt_type": "ET",
28  "inst_id": "GM1015",
29  "is_invoice": true
30 }
```

The response body is a JSON object with the following fields:

```
1 {
2   "tran_ref": "CT0202310040000049501",
3   "tran_date_time": "2023-10-04T14:43:18.000Z",
4   "inst_id": "GM1015",
5   "orgnl_txn_info": {
6     "tran_ref": "CT0202310040000049501",
7     "end_to_end_tran_ref": "20231004SL1012144316"
8   },
9   "status": {
10    "type": "ACCP",
11    "code": "0000",
12    "description": "Success"
13  }
14 }
```

4.3.4.2 Credit Transfer Inbound Request to Core Banking System

```
{
  "tran_ref": "CT0202310040000049501",
  "tran_date_time": "2023-10-04T14:43:18.000Z",
  "receiver": {
    "amount": 500.00,
    "inst_bic": "ICBGGMGS",
    "ctry": "GM",
    "ccy": "GMD",
    "acct_no": "1000000000",
    "inst_id": "GM1015",
    "acct_name": "Mr Papss Sudo-User",
    "addr": "user@papss.com"
  },
  "sender": {
    "amount": 253566.50,
    "inst_bic": "ICBZSLFR",
    "ctry": "SL",
    "ccy": "SLE",
    "acct_no": "435000100",
    "inst_id": "SL1012",
    "acct_name": "John Doe",
    "addr": "Tema"
  },
  "ctgy_purp": "SUPP",
  "narration": "Trade Transfer Invoice for purchase of Cassava",
  "end_to_end_tran_ref": "20231004SL1012144316",
  "lc_instrmt_type": "ET",

```

This item is c

```

    "inst_id": "GM1015",
    "is_invoice": true
  }
}

```

4.3.2.3 Credit Transfer Inbound Response Payload from Core Banking System

```

{
  "tran_ref": "CT0202310040000049501",
  "tran_date_time": "2023-10-04T14:43:18.000Z",
  "inst_id": "GM1015",
  "orgnl_txn_info": {
    "tran_ref": "CT0202310040000049501",
    "end_to_end_tran_ref": "20231004SL1012144316"
  },
  "status": {
    "type": "ACCP",
    "code": "0000",
    "description": "Success"
  }
}

```

4.3.3 Settlement Recon REST API Specification

4.3.3.1 Settlement Recon Inbound Request and Response

The screenshot shows a REST client interface with a POST request to the endpoint `((inboundMockServerBaseUrl))/papss/api/v1/inbound/settlement-recon`. The request body is a JSON object containing transaction details and a large XML payload for settlement reconciliation. The response body is a JSON object indicating a successful transaction.

Request Body (JSON):

```

{
  "tran_ref": "CT0202309180000005902",
  "tran_date_time": "2023-09-18T00:00:29.000Z",
  "inst_id": "GM1015",
  "recon_xml_payload": "<?xml version='1.0' encoding='UTF-8'?><ncrecon xmlns='urn:montran:recon:001'><n\t<settlement instgAgId='GM1015' msgId='CT0202309180000005902' processTime='2023-09-18T00:00:29Z' /><n\t<summary><n\t\t<totalSent amount='0.00' ccy='GMD' count='0' fee='0.00' /><n\t\t<totalReceived amount='0.00' ccy='GMD' count='0' /><n\t\t<total amount='0.00' ccy='GMD' count='0' fee='0.00' /><n\t\t<openingBalance amount='0.00' ccy='GMD' /><n\t\t<closingBalance amount='0.00' ccy='GMD' /></n\t</summary></ncrecon><n",
  "recon_json_payload": "{\n  \"settlement\": {\n    \"instgAgId\": \"GM1015\", \n    \"msgId\": \"CT0202309180000005902\", \n    \"processTime\": \"2023-09-18T00:00:29Z\", \n    \"summary\": {\n      \"totalSent\": {\n        \"amount\": \"0.00\", \n        \"ccy\": \"GMD\", \n        \"count\": \"0\", \n        \"fee\": \"0.00\" \n      }, \n      \"totalReceived\": {\n        \"amount\": \"0.00\", \n        \"ccy\": \"GMD\", \n        \"count\": \"0\" \n      }, \n      \"total\": {\n        \"amount\": \"0.00\", \n        \"ccy\": \"GMD\", \n        \"count\": \"0\", \n        \"fee\": \"0.00\" \n      }, \n      \"openingBalance\": {\n        \"amount\": \"0.00\", \n        \"ccy\": \"GMD\" \n      }, \n      \"closingBalance\": {\n        \"amount\": \"0.00\", \n        \"ccy\": \"GMD\" \n      } \n    } \n  } \n}"
}

```

Response Body (JSON):

```

{
  "tran_ref": "CT0202309180000005902",
  "tran_date_time": "2023-09-18T00:00:29.000Z",
  "inst_id": "GM1015",
  "status": {
    "type": "ACCP",
    "code": "0000",
    "description": "Success."
  }
}

```

4.3.3.2 Settlement Recon Inbound Request to Core Banking System

```

{
  "tran_ref": "CT0202309180000005902",

```

This item is deprecated.

```

    "tran_date_time": "2023-09-18T00:00:29.000Z",
    "inst_id": "GM1015",
    "recon_xml_payload": "<?xml version=\"1.0\" encoding=\"UTF-8\"?>\n<recon
xmlns=\"urn:montran:rcon.001\">\n\t<settlement instgAgtId=\"GM1015\"
msgId=\"CT0202309180000005902\" processTime=\"2023-09-
18T00:00:29Z\"/>\n\t<summary>\n\t\t<totalSent amount=\"0.00\" ccy=\"GMD\" count=\"0\"
fee=\"0.00\"/>\n\t\t<totalReceived amount=\"0.00\" ccy=\"GMD\" count=\"0\"/>\n\t\t<total
amount=\"0.00\" ccy=\"GMD\" count=\"0\" fee=\"0.00\"/>\n\t\t<openingBalance amount=\"0.00\"
ccy=\"GMD\"/>\n\t\t<closingBalance amount=\"0.00\" ccy=\"GMD\"/>\n\t</summary>\n</recon>\n",
    "recon_json_payload":
    "{ \"settlement\": { \"instgAgtId\": \"GM1015\", \"msgId\": \"CT0202309180000005902\", \"processTime\": \"2023-09-
18T00:00:29Z\" }, \"summary\": { \"totalSent\": { \"amount\": \"0.00\", \"ccy\": \"GMD\", \"count\": \"0\"
\", \"fee\": \"0.00\" }, \"totalReceived\": { \"amount\": \"0.00\", \"ccy\": \"GMD\", \"count\": \"0\" }, \"
total\": { \"amount\": \"0.00\", \"ccy\": \"GMD\", \"count\": \"0\", \"fee\": \"0.00\" }, \"openingBalanc
e\": { \"amount\": \"0.00\", \"ccy\": \"GMD\" }, \"closingBalance\": { \"amount\": \"0.00\", \"ccy\": \"GM
D\" } } }" }

```

4.3.3.3 Settlement Recon Inbound Response from Core Banking System

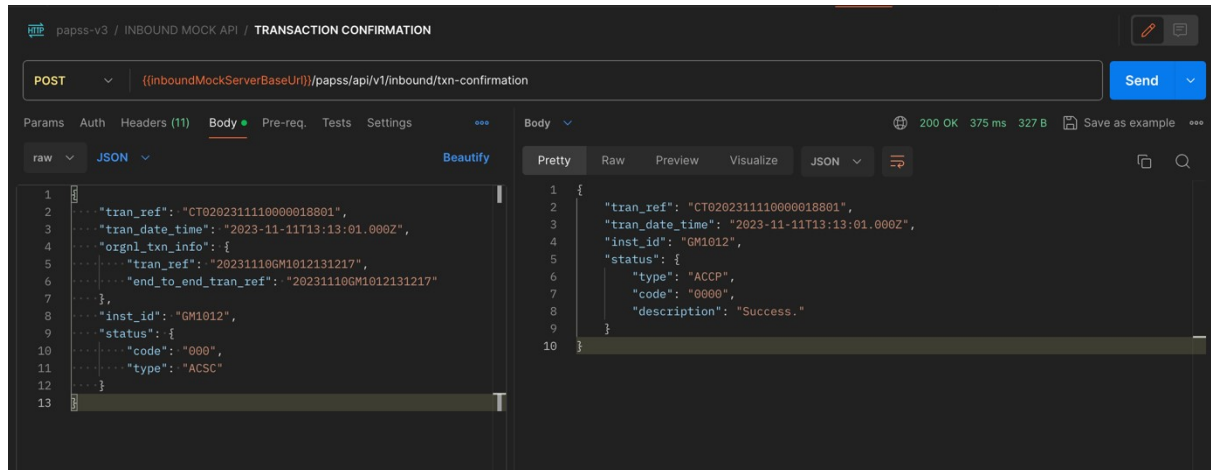
```

{
    "tran_ref": "CT0202309180000005902",
    "tran_date_time": "2023-09-18T00:00:29.000Z",
    "inst_id": "GM1015",
    "status": {
        "type": "ACCP",
        "code": "0000",
        "description": "Success."
    }
}

```

4.3.4 Transaction Confirmation REST API Specification

4.3.4.1 Transaction Confirmation Request & Response



4.3.4.2 Transaction Confirmation Inbound Request to Core Banking System

```
{
  "tran_ref": "CT0202311110000018801",
  "tran_date_time": "2023-11-11T13:13:01.000Z",
  "orgnl_txn_info": {
    "tran_ref": "20231110GM1012131217",
    "end_to_end_tran_ref": "20231110GM1012131217"
  },
  "inst_id": "GM1012",
  "status": {
    "code": "000",
    "type": "ACSC"
  }
}
```

4.3.4.3 Transaction Confirmation Inbound Response to Core Banking System

```
{
  "tran_ref": "CT0202311110000018801",
  "tran_date_time": "2023-11-11T13:13:01.000Z",
  "inst_id": "GM1012",
  "status": {
    "type": "ACCP",
    "code": "0000",
    "description": "Success."
  }
}
```

5 Annexure

Annexure 1: PAPSS Response Codes

This item is c

Value	Mapping	Description
-------	---------	-------------

0	""	"Success."
100	"MS03"	"Internal processing error."
500	"AB05/TM01"	"Timeout error. The code reported at sender is AB05 and the coder reported at receiver is TM01."
501	"MS02"	"Payment rejected by receiver Participant."
502	"AB08"	"Receiver Participant is disconnected."
1000	"MS03"	"Generic error."
1001	"MS03"	"Generic validation error."
1002	"FF01"	"The currency is invalid."
1003	"FF01"	"The amount is invalid (less than or equal to zero)."
1004	"DNOR"	"Value of field DbtrAgt is invalid: No active Participant with this BIC was found. The value is different than InstgAgt. The value is different than the sender Participant detected at the communication channel level."
1005	"CNOR"	"Value of field CdtrAgt is invalid: No active Participant with this BIC was found."
1006	"RC01"	"Value of field InstgAgt is invalid: No active Participant with this BIC was found. The value is different than the sender Participant detected at the communication channel level."
1007	"RC01"	"Unused"
1008	"AC01"	"The debtor IBAN is invalid."
1009	"AC01"	"The creditor IBAN is invalid."
1010	"AM05"	"A payment with the same reference has been detected within processing."
1011	"AM05"	"A payment with the same reference has been detected as already processed."
1012	"FF01"	"Settlement date is invalid (IntrBkSttImDt)."
1013	"FF01"	"Invalid XML format."
1014	""	"Unused."
1015	"AB05"	"AB05", "Invalid time – the value of field Acceptance Date Time is older than the current time with the timeout parameter value configured in the payment schema."
1016	"AG09"	"Invalid original reference – during the processing of a pacs.028 message, no original transaction with that reference was found to have been processed within the last 24 hours."
1017	"AG09"	"Invalid status of the original transaction – during the processing of a pacs.002/pacs.028, the referred transaction has no appropriate status (message: WAIT_RECEIVER, transaction: HOLD)."
1018	"RC01"	"No valid sender – the sender Participant is not ACTIVE."
1019	"AG01"	"The payment schema is not defined for the currency mentioned in the message."
1020	"AM02"	"The amount is too large in comparison to the parameter defined in the payment schema."
1021	"AG10"	"The sender is BLOCKED (temporary deactivated)."
1022	"RR04"	"The Participant is not mapped in the payment schema. Both Participants mentioned in the message need to be checked."
1026	"FF01"	"Invalid XML field. It is setup if the validation of some fields from the XML message do not respect the validation rules that are not included in the XML schema, e.g.: ClrSys, SvcLvl, presence InstdAgt, value NbOfTx. Message not XML, Reference not valid, 2 cases, reference not start with date or correct date, not followed by PAPSS ID of sender"
1027	"RC01"	"Invalid receiver message field. It could be the AppHdr.To.FiId.FinInstId.BICFI or the Assignee (for camt.* messages)."
1028	"FF01"	"The value of field Original Settlement Date is not valid in comparison to the payment schema and the current day."
1029	"RC01"	"Transaction/message On Us. Debtor Agent and Creditor Agent are equal."
1030	"RC01"	"Invalid originator. The sender of the payment is not the instigated agent."
1031	"MS03"	"Invalid national payment message. In case of an escrow reservation the sender country must be the same as the receiver country."
1032	"MS03"	"Missing sender Settlement Region. No settlement region was found for the sender country and currency."
1033	"MS03"	"Missing instructed amount."
1034	"MS05"	"Missing receiver Settlement Region."
1035	"MS03"	"Missing FX Rate."
1036	"MS05"	"Account disabled."
1037	"MS03"	"Invalid sender agent."
1038	"MS03"	"Compliance check failure."
2000	"AM23"	"Insufficient funds for the transaction clearing."
3000	"FF01"	"The Participant has no ACTIVE registered DS certificate for the validation of messages."
3001	"FF01"	"The XML message does not entail the digital signature in the specified format."
3002	"FF01"	"The digital signature does not protect the entire XML message."
3003	"FF01"	"Invalid digital signature."
3004	"FF01"	"The certificate used for the signature is expired or it has been revoked."
9001	"MS03"	"Settlement posting failure."

Annexure 2: PAPSS Integration Test Cases

This item is c

TEST SCRIPT FOR PAPSS SSA1 and SSA3 TRANSACTIONS						
Test ID	Module	Description	Testing Steps	Expected Result/Outcome	Account Entries	Status -

						Pass/Fail
TC1	Account enquiry Outward (Valid Account)	Account enquiry	Input the recipient account details correctly	The receiving participant gets a response "True"		
TC2	Account enquiry Outward (Invalid Account)	Account enquiry	Input the recipient account details wrongly	The receiving participant gets a response "False"		
TC3	Account enquiry Inward (Valid Account)	Account enquiry	Input the recipient account details correctly	Positive validation of the account number		
TC4	Account enquiry Inward (Invalid Account)	Account enquiry	Input the recipient account details wrongly	Negative response received		
TC5	Rate enquiry for Direct Payment	Rate Enquiry	Carry out enquiry on beneficiary's amount	Beneficiary amount is displayed - source currency, destination currency and USD		
TC6	Direct Payment: Outward (to a region on Instant payment)	Payment	Send a transaction using Sender currency to a beneficiary account on an instant payment region	Successful ACSC for Instant Payment Region		
TC7	Direct Payment: Outward (to a region on non Instant payment)	Payment	Send a transaction using Sender currency to a beneficiary account on non instant payment region	Successful ACSP for non-Instant Payment Region		
TC8	Direct Payment: Outward (using customer account with insufficient balance)	Payment	Send transaction using a customer account with an insufficient balance	Rejected transaction		
TC9	Direct Payment: Inward (from a region on Instant payment)	Payment	Receive payment from an Instant payment region	Successful ACSC for Instant Payment Region		
TC10	Direct Payment: Inward (from a region on Non-Instant payment)	Payment	Receive payment from a non-instant payment region	Successful ACSP for non-Instant Payment Region		
TC11	Direct Payment: Outward (Testing fees in T4)	Payment	Send transaction using a customer account with an amount checking for fee in T4	Successful fee charged for that Tier of payment		
TC12	Rate enquiry for Invoice Payment	Rate Enquiry	Carry out enquiry on sender's amount	Sender's amount is displayed		
TC13	Invoice Payment: Outward (to a region on Instant payment)	Payment	Send a transaction using Sender currency to a beneficiary account on instant payment	Successful ACSC for Instant Payment Region		
TC14	Invoice Payment: Outward (to a region on Non Instant payment)	Payment	Send a transaction using Sender currency to a beneficiary account on non instant payment	Successful ACSP for non-Instant Payment Region		
TC15	Invoice Payment: Inward (from a region on Instant payment)	Payment	Receive payment from an Instant payment region	Successful ACSC for Instant Payment Region		
TC16	Invoice Payment: Inward (from a region on Non-instant payment)	Payment	Receive payment from a non-instant payment region	Successful ACSP for non-Instant Payment Region		
TC17	Invoice Payment: Outward (Testing fees in T2)	Payment	Send transaction with an amount checking for fee in T2	Successful fee charged for that Tier of payment		
TC18	PAPSS Participant List Query	Report	Query PAPSS for the list of all participants	Participants list is displayed (online,offline, active and inactive)		

This item is classified as P rivate

TC1 9	Status Query for transactions	Status Query	Look for the previous transactions and get their status	Status of previous transactions are displayed		
TC2 0	Final Status Message (Non-Instant Transaction)	Message	Confirm receipt of the status message (T+1) from effective transaction date to and from a noninstant region. This is for both Sender & Receiver	Status message with ACSC is received		
Below Test scenarios are to be done on the Production environment only						
TC2 1	RTGS: Funding of PAPSS Clearing Account	Funding	Fund PAPSS local RTGS account	Confirmation of MT910 message and crediting of participant's clearing account in PAPSS (this is to be done on Production only)		
TC2 2	RTGS: Defunding of PAPSS Clearing Account	Defunding	Initiate a request to defund from Clearing account using Manual Fund Release	Confirmation of credit on participant's RTGS account. To be done on Production environment only.		
Payments in LCY						
TC2 3	Direct Payment: Outward (to an SSA3 participant on Instant Payment Region) T1 transaction	Payment - SSA3 to SSA3 Accounts	Send SSA3 transaction using sender's currency to an SSA3 beneficiary account on an instant payment region. Check fees for T1 transaction	Successful ACSC received and transaction completed for Instant Payment Region. NOSTRO accounts of sender and receiver, debited and credited.	Confirm that the sending account was debited to credit PAPSS GL/Transit	
TC2 4	Direct Payment: Outward (to an SSA3 participant on NonInstant Payment Region)	Payment - SSA3 to SSA3 Accounts	Send SSA3 transaction using sender's currency to an SSA3 beneficiary account on a noninstant payment region	Successful ACSP received and transaction is on postponed status for Non-Instant Payment Region. NOSTRO accounts of sender and receiver, debited and credited.	Confirm that the amount is held to credit at T+1	
TC2 5	Direct Payment: Outward (to an SSA1 participant on Instant Payment Region) T3 transaction	Payment - SSA3 to SSA1 Accounts	Send SSA3 transaction using sender's currency to an SSA1 beneficiary account on an instant payment region. Check fees for T3 transaction	Successful ACSC received and transaction completed for Instant Payment Region. Sender's NOSTRO account debited and Receiver's CA credited.	Confirm that the sending account was debited to credit PAPSS GL/Transit	
TC2 6	Direct Payment: Outward (to an SSA1 participant on NonInstant Payment Region)	Payment - SSA3 to SSA1 Accounts	Send SSA3 transaction using sender's currency to an SSA1 beneficiary account on a noninstant payment region	Successful ACSP received and transaction is on postponed status for Non-Instant Payment Region. Sender's NOSTRO account debited and Receiver's CA credited.	Confirm that the amount is held to credit at T+1	
TC2 7	Direct Payment: Outward (using customer account with insufficient balance)	Payment - SSA3 to SSA3 Accounts	Send SSA3 transaction using a customer account with an insufficient balance	Transaction Rejected		
TC2 8	Direct Payment: Inward (from an SSA3 participant on Instant Payment Region)	Payment - SSA3 to SSA3 Accounts	Receive SSA3 transaction from an instant payment region	Successful ACSC received and transaction completed for Instant Payment Region. NOSTRO accounts of sender and receiver debited and credited.	Confirm that the PAPSS GL/Transit is debited and receiver's account credited.	
TC2 9	Direct Payment: Inward (from an SSA3 participant on NonInstant Payment Region)	Payment - SSA3 to SSA3 Accounts	Receive SSA3 payment from a noninstant payment region	Successful ACSP received and transaction is on postponed status for Non-Instant Payment Region. NOSTRO accounts of sender and receiver debited and credited.	Confirmation of inwards transaction in database, then credit the beneficiary's account at T+1	

This item is classified as Private

TC3 0	Rate enquiry for Invoice Payment	Rate Enquiry	Carry out enquiry on sender's amount	Sender's amount is displayed		
TC3 1	Invoice Payment: Outward (to an SSA3)	Payment - SSA3 to SSA3 Accounts	Send a transaction using Receiver's	Successful ACSC received and transaction completed for Instant	Confirm that the sending	

	participant on Instant Payment Region) T2 transaction		currency on SSA3 to an instant payment region. Check fees for T2 transaction	Payment Region. Sender's and receiver's NOSTRO accounts debited and credited.	account was debited to credit PAPSS GL/Transit	
TC3 2	Invoice Payment: Outward (to an SSA3 participant on Non Instant Payment Region)	Payment - SSA3 to SSA3 Accounts	Send a transaction using Receiver's currency on SSA3 to a non-instant payment region.	Successful ACSP received and transaction postponed for nonInstant Payment Region. Sender's and receiver's NOSTRO accounts debited and credited.	Confirm that the amount is held to credit at T+1	
TC3 3	Invoice Payment: Outward (to an SSA1 participant on Instant Payment Region)	Payment - SSA3 to SSA1 Accounts	Send a transaction using Receiver's currency on SSA1 to an instant payment region.	Successful ACSC received and transaction completed for Instant Payment Region. Sender's NOSTRO account debited and Receiver's CA credited.	Confirm that the sending account was debited to credit PAPSS GL/Transit	
TC3 4	Invoice Payment: Outward (to an SSA1 participant on Non Instant Payment Region)	Payment - SSA3 to SSA1 Accounts	Send a transaction using Receiver's currency on SSA1 to a non-instant payment region.	Successful ACSP received and transaction postponed for a NonInstant Payment Region. Sender's NOSTRO account debited and Receiver's CA credited.	Confirm that the amount is held to credit at T+1	
TC3 5	Invoice Payment: Inward (from an SSA1 participant on Instant Payment Region)	Payment - SSA1 to SSA3 Accounts	Receive a transaction from an SSA1 participant on instant payment region.	Successful ACSC received and transaction completed for Instant Payment Region. Receiver's CA debited and Sender's NOSTRO account credited.	Confirm that the PAPSS GL/Transit is debited and receiver's account credited.	
TC3 6	Invoice Payment: Inwards (from an SSA1 participant on Non Instant Payment Region)	Payment - SSA1 to SSA3 Accounts	Receive a transaction from an SSA1 participant on noninstant payment region.	Successful ACSP received and transaction postponed for NonInstant Payment Region. Receiver's CA debited and Sender's NOSTRO account credited.	Confirmation of inwards transaction in database, then credit the beneficiary's account at T+1	

Payments in USD (SSA3 Participants Only)

TC3 7	Send Payment to an Instant Region	Payment	Send transaction to an instant region in USD with an amount checking for fee in T2	Successful ACSC received and transaction completed for instant region. The SA3 accounts of the participants are debited and credited. Successful fee charged for that Tier of payment.	Confirm that the USD dorm account is debited to credit a USD PAPSS GL	
TC3 8	Receive Payment from an Instant Region	Payment	Receive transaction in USD, from an instant region	Successful ACSC received and transaction completed for an instant region payment. The SA3 accounts of the participants are debited and credited.	Confirm that the USD PAPSS GL is debited to credit the receiver's dorm account	
TC3 9	Send Payment to a Non-Instant Region	Payment	Send transaction to a non-instant region in USD with an amount checking for fee in T1	Successful ACSP received and transaction postponed for a noninstant region payment. The SA3 accounts of the participants are impacted. Successful fee charged for that Tier of payment	Confirm that the USD amount is held to credit at T+1	
TC4 0	Receive Payment from a Non-Instant Region	Payment	Receive transaction from a non-instant region in USD	Successful ACSP received and transaction postponed for a noninstant region payment. The SA3 accounts of the participants are impacted.	Confirmation of inwards transaction in database, then credit the beneficiary's dorm account at T+1	

Accounts Validation Tests

TC4 1	Send payment to a USD Account for a LCY Payment	Payment	Send payment to a USD Account for a LCY Payment	Rejected transaction		
----------	---	---------	---	----------------------	--	--

This item is classified as Private

TC4 2	Receive payment from a USD Account for a LCY Payment	Payment	Receive payment from a USD Account for a LCY Payment	Rejected transaction		
TC4 3	Send payment to a LCY Account for a USD Payment	Payment	Send payment to a LCY Account for a USD Payment	Rejected transaction		
TC4 4	Receive payment from a LCY Account for a USD Payment	Payment	Receive payment from a LCY Account for a USD Payment	Rejected transaction		
Transaction Status Queries and Confirmations						
TC4 5	PAPSS Participant List Query	Report	Query PAPSS for the list of all participants	Participants list is displayed (online,offline, active and inactive)		
TC4 6	Status Query for transactions	Status Query	Look for the previous transactions and get their status	Status of previous transactions are displayed		
TC4 7	Final Status Message (Non-Instant Transaction)	Message	Confirm receipt of the status message (T+1) from effective transaction date to and from a noninstant region. This is for both Sender & Receiver	Status message with ACSC is received		
Below Test scenarios are to be done on the Production environment only						
TC4 8	SA3 Funding in USD at Afrexim	Funding	Fund Settlement Account in USD	Participant's Dynamic Limit reflects the USD funding done. SA3 Accounts Position reflects the amount in USD		
TC4 9	Set FX Rate	FX Rate	Set FX Rate for the SSA3 Payments	The NOSTRO account reflects the local currency equivalent of the USD Funding.		

Annexure 3: PAPSS Schema Codes

Payment Recall (Camt.056) Reason Codes:

- CUTA - Cancel Upon Unable To Apply (CUTA)
- COVR - Cover Cancelled Or Returned (COVR)
- DUPL - Duplicate Payment (DUPL) • FRAD - Fraudulent Origin (FRAD)
- CURR - Incorrect Currency (CURR) • AGNT - Incorrect Agent (AGNT)
- AM09 - Incorrect Amount (AM09)
- NARR - Narrative (NARR)
- CUST - Requested By Customer (CUST)
- TECH - Technical Problem (TECH)
- UPAY - Undue Payment(UPAY)

Payment Recall NACK (Camt.029) Reason Codes:

- FRAD - Fraudulent originated credit transfer.
- CUST - Customer Decision, Beneficiary does not want to honour the Recall.
- ARDT - Already returned/recalled transaction.
- AC04 - Closed Account Number
- AM04 - Insufficient funds on the account.
- NOAS - No Answer From Customer
- NOOR - No Original Transaction Received

